



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: / /

Laboratory Manual

Academic Year	:	2024-25	Semester	:	4
Course code	:	CSE206	Course name	:	DATABASE MANAGEMENT SYSTEM

Practical - 11

Aim: -

Your organization maintains an employee database that includes details like names and ages. To ensure data integrity and audit compliance, any record deleted due to specific conditions must first be archived. You are tasked with implementing a system where records of employees aged 21 are deleted from the main table but saved in an archive table before deletion.

Tasks:-

The HR department wants to remove all records of employees aged 21 from the Employee table as part of a policy update. Use a PL/SQL block to delete records from the main table based on the condition.

```

-- Step 1: Create Employee Table
CREATE TABLE Employee (
  emp_id NUMBER PRIMARY KEY,
  name VARCHAR2(100),
  age NUMBER
);

-- Step 2: Create Employee_Archive Table
CREATE TABLE Employee_Archive (
  emp_id NUMBER PRIMARY KEY,

```

Script Output: Task completed in 0.07 seconds

Table EMPLOYEE created.

The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS23.sql'. The 'Query Builder' tab is active, displaying the following SQL script:

```
name VARCHAR2(100),
age NUMBER
);

-- Step 2: Create Employee_Archive Table
CREATE TABLE Employee_Archive (
  emp_id NUMBER PRIMARY KEY,
  name VARCHAR2(100),
  age NUMBER,
  deleted_at TIMESTAMP DEFAULT SYSTIMESTAMP
);

-- Step 3: Insert Sample Data
```

The 'Script Output' tab shows the execution results:

```
Table EMPLOYEE created.

Table EMPLOYEE_ARCHIVE created.
```

The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS43.sql'. The 'Query Builder' tab is active, displaying the following SQL script:

```
deleted_at TIMESTAMP DEFAULT SYSTIMESTAMP
);

-- Step 3: Insert Sample Data
INSERT INTO Employee VALUES (1, 'John Doe', 21);
INSERT INTO Employee VALUES (2, 'Jane Smith', 25);
INSERT INTO Employee VALUES (3, 'Alice Brown', 21);
INSERT INTO Employee VALUES (4, 'Bob White', 30);
COMMIT;

-- Step 4: Create Trigger for Archiving Before Deletion
CREATE OR REPLACE TRIGGER trg_archive_before_delete
BEFORE DELETE ON Employee
```

The 'Script Output' tab shows the execution results:

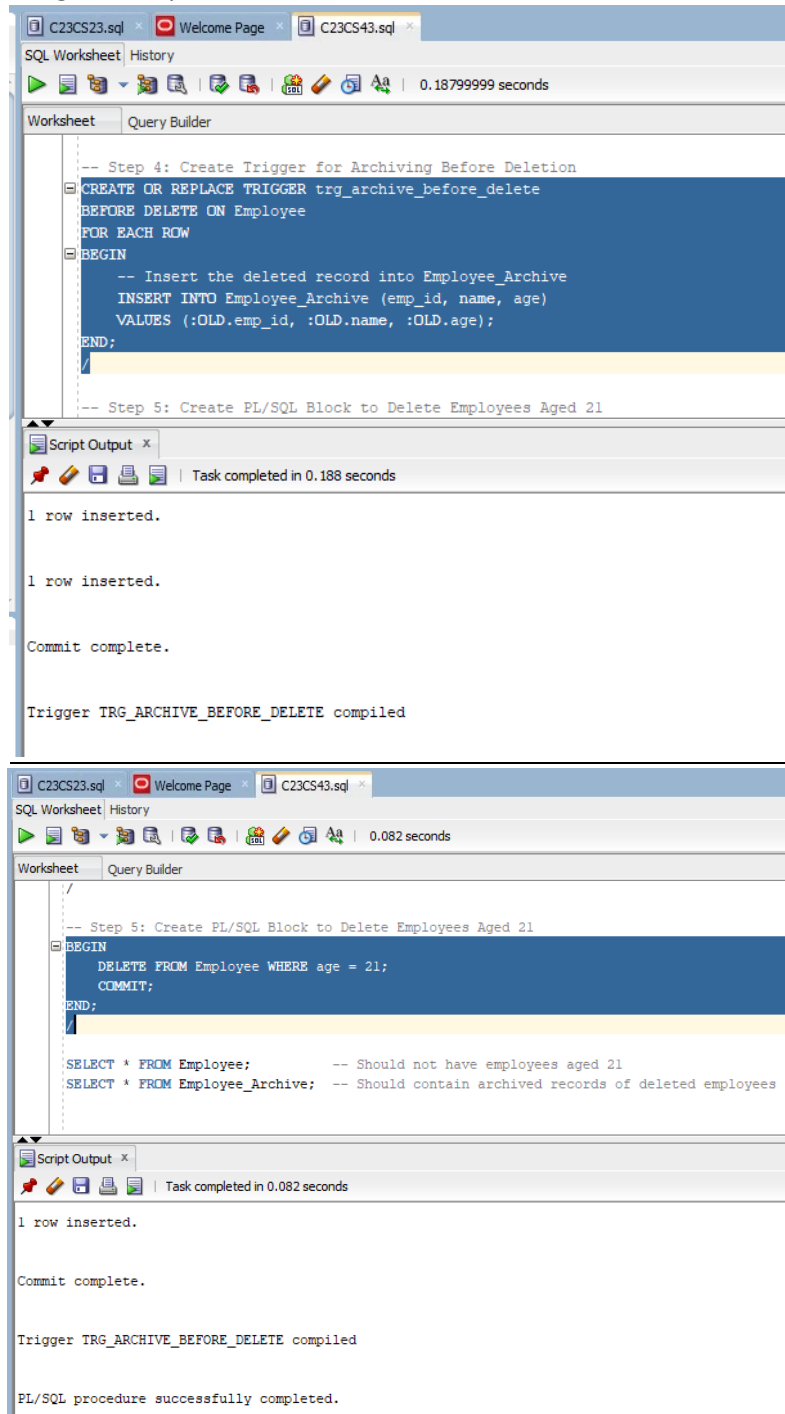
```
1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.
```

2. Implement a trigger that automatically archives the original record in a separate table before deletion



The screenshot displays the SQL Developer interface with two tabs: C23CS23.sql and C23CS43.sql. The active tab, C23CS43.sql, contains a PL/SQL script. The script is divided into two sections: Step 4 and Step 5. Step 4 involves creating a trigger named trg_archive_before_delete that archives records before deletion. Step 5 involves creating a PL/SQL block to delete employees aged 21. The script is executed, and the output shows successful completion of both steps.

```
-- Step 4: Create Trigger for Archiving Before Deletion
CREATE OR REPLACE TRIGGER trg_archive_before_delete
BEFORE DELETE ON Employee
FOR EACH ROW
BEGIN
    -- Insert the deleted record into Employee_Archive
    INSERT INTO Employee_Archive (emp_id, name, age)
    VALUES (:OLD.emp_id, :OLD.name, :OLD.age);
END;
```

-- Step 5: Create PL/SQL Block to Delete Employees Aged 21

```
BEGIN
    DELETE FROM Employee WHERE age = 21;
    COMMIT;
END;
```

Script Output x

Task completed in 0.188 seconds

1 row inserted.

1 row inserted.

Commit complete.

Trigger TRG_ARCHIVE_BEFORE_DELETE compiled

The second screenshot shows the same SQL Developer interface with the C23CS43.sql tab. The script is the same as in the first screenshot, but the output shows the successful completion of the PL/SQL procedure. The output also includes a confirmation message: "PL/SQL procedure successfully completed."

```
-- Step 5: Create PL/SQL Block to Delete Employees Aged 21
BEGIN
    DELETE FROM Employee WHERE age = 21;
    COMMIT;
END;
```

Script Output x

Task completed in 0.082 seconds

1 row inserted.

Commit complete.

Trigger TRG_ARCHIVE_BEFORE_DELETE compiled

PL/SQL procedure successfully completed.

The screenshot shows the SQL Developer interface with a worksheet containing a PL/SQL block. The block is designed to delete employees aged 21 and then verify the deletion by querying the Employee and Employee_Archive tables. The execution results show two rows in the Employee table, indicating that the deletion was successful for the specified age group.

```
-- Step 5: Create PL/SQL Block to Delete Employees Aged 21
BEGIN
    DELETE FROM Employee WHERE age = 21;
    COMMIT;
END;
/

SELECT * FROM Employee; -- Should not have employees aged 21
SELECT * FROM Employee_Archive; -- Should contain archived records of deleted employees
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.063 seconds

EMP_ID	NAME	AGE
1	2 Jane Smith	25
2	4 Bob White	30

The screenshot shows the SQL Developer interface with a worksheet containing a PL/SQL block. The block is designed to delete employees aged 21 and then verify the deletion by querying the Employee and Employee_Archive tables. The execution results show two rows in the Employee table, indicating that the deletion was successful for the specified age group.

```
-- Step 5: Create PL/SQL Block to Delete Employees Aged 21
BEGIN
    DELETE FROM Employee WHERE age = 21;
    COMMIT;
END;
/

SELECT * FROM Employee; -- Should not have employees aged 21
SELECT * FROM Employee_Archive; -- Should contain archived records of deleted employees
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.062 seconds

EMP_ID	NAME	AGE	DELETED_AT
1	1 John Doe	21	27-FEB-25 09.35.22.744000000 AM
2	3 Alice Brown	21	27-FEB-25 09.35.22.744000000 AM